

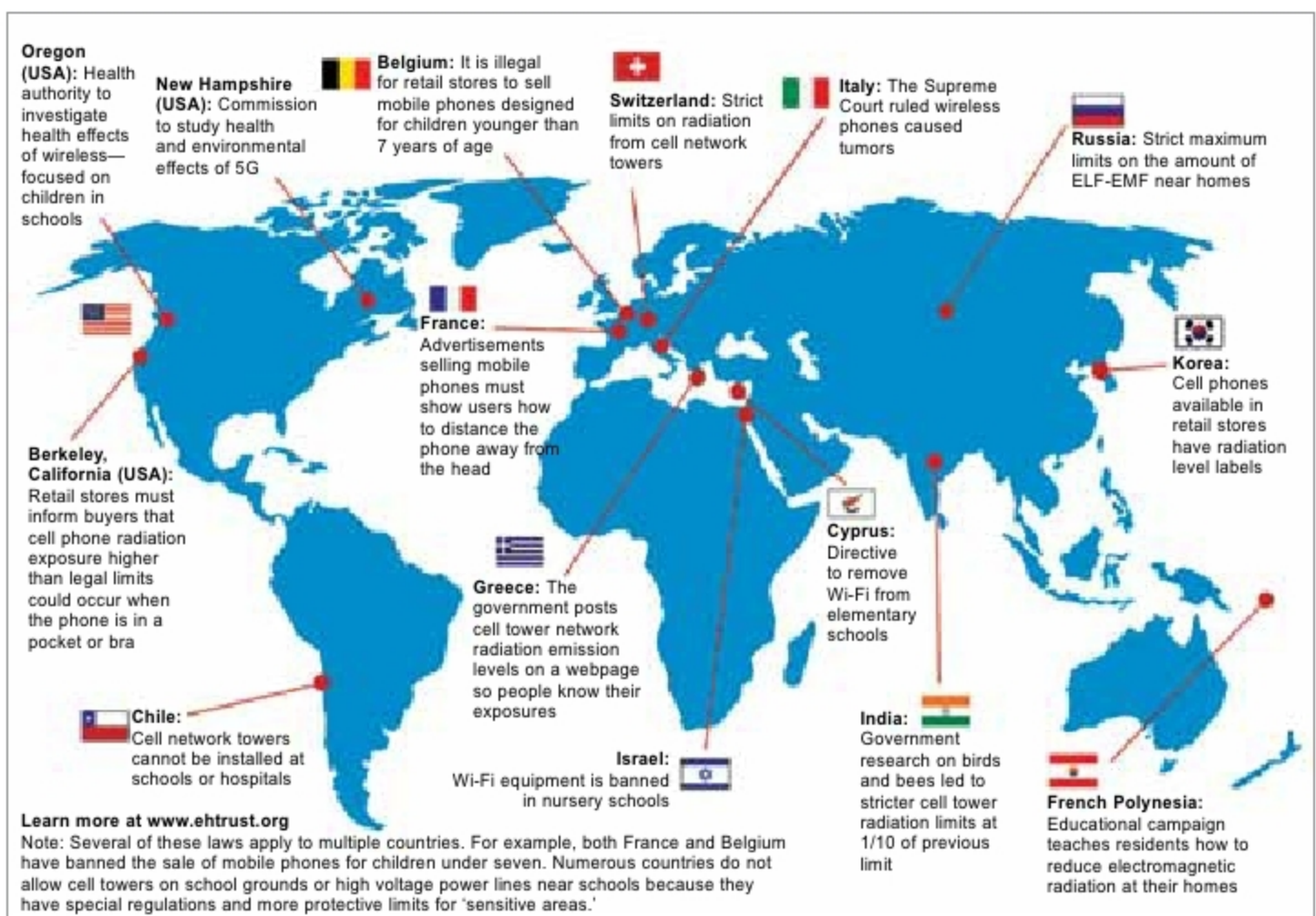
Cellphones. Our cellphones go with us wherever we go, thereby causing maximum exposure to radiation. The exposure is higher if the mobile being used is older or closer to the body, which is generally the case. On doubling the distance from the source, the radiation gets almost four times weaker. But most of the radiation is the result of using Wi-Fi (2.4GHz or 5GHz) and mobile data (1-2GHz for 4G) to get access to the Internet.

Laptops. Laptops have gained more importance in homes due to their portability over computers. Much like cellphones, radiation from laptops is highest when it is connected to Wi-Fi or Bluetooth for browsing the Internet or sharing files in a close range, respectively. It is recommended that there must be a minimum distance of 20cm between the laptop and the user.

Headphones. Headphones are attached to our ears and head when we use them. Wireless headsets emit RF signals to communicate with phones. While wired earphones increase the risks to the brain since the phone is close to the body, the metal in the wire also amplifies the radiation.

Wireless routers. Most homes have Wi-Fi routers installed, which when transmitting data, expose nearby areas to RF signals. However, the amount of radiation absorbed by the human body from a wireless router is quite less as compared to that transmitted from using a cellphone. If the obstructions in the path are of higher thickness, radio waves may be absorbed completely.

Tablets. Tablets are quite popular due to their compact nature and range of functionalities available, especially among kids for purposes like studying, playing games, and practising digital drawings. These devices are connected via Wi-Fi, Bluetooth, or a cellular connection to allow Internet connectivity. However, these emit radiations if the Internet connection is turned on even when the screen is off.



Worldwide policies for wireless devices (Credit: <https://ehtrust.org>)

Wearables. Wearables such as smartwatches and fitness trackers are worn on the body and operated via connectivity to other wireless devices like smartphones and laptops. These continuously emit radiation, and hence we are exposed to RF energy 24/7.

Others. Other devices like mouse and keyboards also emit radiations if these utilise wireless connectivity to find the computer they want to work with. Even for wireless microphones, the radiation is high when it is being used to send a large amount of information.

With the growing number of Internet of Things (IoT) connected devices, the sources of RF are bound to grow. This includes smart refrigerators, smart meters, baby monitors, and so on.

What can be done?

Scientists have been studying the effects of long-term exposure to various levels of RF for many years, but it is a tricky assessment. This is because, in such a long time, there are numerous factors that cannot be controlled but may impact the inference. Radiation from RF energy is non-ionising in nature. Due to this, most of them agree that there is no proof of these being carcinogenic in the long run, but it has not been ruled out. So, it has to

be made sure that everyone can use the devices safely.

Government agencies set safety guidelines and regulations to limit RF exposure from wireless devices and protect consumers. But there are no international standards, so the permissible exposure limits vary in different nations. The restrictive limits on whole-body exposure mainly lie in the frequency range of 30-300MHz since the entire human body can efficiently absorb RF waves.

EMF meters are used to measure electromagnetic radiations, with more expensive ones having the option of customisation in the high-frequency range.

Users can take some measures to avoid problems in the long run. They can make sure to place their laptop on a table to maintain a safe distance during work, especially when connected to Wi-Fi. By switching off wireless routers when not in use or removing wearables when possible, consumers can ensure that they are not exposed to radiations round the clock.

There are a number of products made of materials like aluminium available in the market for shielding too. For example, some mobile cases can prevent radiation from getting absorbed by the body. **EFY**